Лабораторная работа №14

Статическая маршрутизация в Интернете. Настройка

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1 Цель работы

Настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи.

2 Выполнение работы

Первым делом нам нужно настроить линку между площадками. Для этого настроим интерфейсы у коммутатора `provider-agko-sw-1`, маршрутизатора `msk-donskaya-agko-gw-1`, маршрутизатора `msk-q42-agko-gw-1`, коммутатора `sch-sochi-agko-sw-1` и маршрутизатора `sch-sochi-agko-gw-1` (рис. #fig:002 – #fig:008):

```
provider-agko-sw-1>enable
Password:
provider-agko-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-agko-sw-1(config)#interface f0/3
provider-agko-sw-1(config-if)#switchport mode trunk

provider-agko-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
exit
provider-agko-sw-1(config)#interface f0/4
provider-agko-sw-1(config-if)#switchport mode trunk
provider-agko-sw-1(config-if)#exit
provider-agko-sw-1(config)#vlan 5
provider-agko-sw-1(config-vlan)#name q42
provider-agko-sw-1(config-vlan)#exit
provider-agko-sw-1(config)#interface vlan 5
provider-agko-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan5, changed state to up
no shutdown
provider-agko-sw-1(config-if)#exit
provider-agko-sw-1(config)#vlan 6
provider-agko-sw-1(config-vlan)#name sochi
provider-agko-sw-1(config-vlan)#exit
provider-agko-sw-1(config)#interface vlan6
provider-agko-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up
no shutdown
provider-agko-sw-1(config-if)#exit
provider-agko-sw-1(config)#^Z
provider-agko-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
provider-agko-sw-1#
```

Рисунок 2.1: Настройка интерфейсов коммутатора provider-agko-sw-1

```
msk-donskaya-agko-gw-1>enable
Password:
msk-donskaya-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-agko-gw-1(config)#enable
% Incomplete command.
msk-donskaya-agko-gw-1(config)#interface f0/1.5
msk-donskaya-agko-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up
encapsulation dot1Q 5
msk-donskaya-agko-gw-1(config-subif)#ip address 10.128.255.1 255.255.255.252
msk-donskaya-agko-gw-1(config-subif)#description q42
msk-donskaya-agko-gw-1(config-subif)#exit
msk-donskaya-agko-gw-1(config)#interface f0/1.6
msk-donskaya-agko-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.6, changed state to up
encapsulation dot1Q 6
msk-donskaya-agko-gw-1(config-subif)#ip address 10.128.255.5 255.255.255.252
msk-donskaya-agko-gw-1(config-subif)#description sochi
msk-donskaya-agko-gw-1(config-subif)#exit
msk-donskaya-agko-gw-1(config)#exit
msk-donskaya-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-donskaya-agko-gw-1#
```

Рисунок 2.2: Настройка интерфейсов маршрутизатора msk-donskaya-agko-gw-1

```
msk-q42-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-agko-gw-1(config)#interface f0/1
msk-q42-agko-gw-1(config-if)#no shutdown

msk-q42-agko-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
msk-q42-agko-gw-1(config)#interface f0/1.5
msk-q42-agko-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up
encapsulation dot1Q5
                                                    ^
% Invalid input detected at '^' marker.

msk-q42-agko-gw-1(config-subif)#encapsulation dot1Q5
                                                    ^
% Invalid input detected at '^' marker.

msk-q42-agko-gw-1(config-subif)#encapsulation dot1Q 5
msk-q42-agko-gw-1(config-subif)#ip address 10.128.255.2 255.255.255.252
msk-q42-agko-gw-1(config-subif)#description donskaya
msk-q42-agko-gw-1(config-subif)#exit
msk-q42-agko-gw-1(config)#exit
msk-q42-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
```

Рисунок 2.3: Настройка интерфейсов маршрутизатора msk-q42-agko-gw-1

```
msk-q42-agko-gw-1#
msk-q42-agko-gw-1#ping 10.128.255.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.128.255.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/3/11 ms

msk-q42-agko-gw-1#
```

Рисунок 2.4: Выполнение проверки

```
User Access Verification

Password:

sch-sochi-agko-sw-1>enable
Password:
sch-sochi-agko-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-agko-sw-1(config)#interface f0/23
sch-sochi-agko-sw-1(config-if)#switchport mode trunk
sch-sochi-agko-sw-1(config-if)#exit
sch-sochi-agko-sw-1(config)#interface f0/24
sch-sochi-agko-sw-1(config-if)#switchport mode trunk
sch-sochi-agko-sw-1(config-if)#exit
sch-sochi-agko-sw-1(config)#vlan 6
sch-sochi-agko-sw-1(config-vlan)#name sochi
sch-sochi-agko-sw-1(config-vlan)#exit
sch-sochi-agko-sw-1(config)#interface vlan6
sch-sochi-agko-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up
no shutdown
sch-sochi-agko-sw-1(config-if)#exit
sch-sochi-agko-sw-1(config)#^Z
sch-sochi-agko-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
sch-sochi-agko-sw-1#
```

Рисунок 2.5: Настройка интерфейсов коммутатора sch-sochi-agko-sw-1

```
User Access Verification

Password:

sch-sochi-agko-gw-1>enable
Password:
sch-sochi-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-agko-gw-1(config)#interface 0/0
                                      ^
% Invalid input detected at '^' marker.

sch-sochi-agko-gw-1(config)#interface f0/0
sch-sochi-agko-gw-1(config-if)#no shutdown

sch-sochi-agko-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
sch-sochi-agko-gw-1(config)#interface f0/0.6
sch-sochi-agko-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.6, changed state to up
encapsulation dot1Q 6
sch-sochi-agko-gw-1(config-subif)#ip address 10.128.255.6 255.255.255.252
sch-sochi-agko-gw-1(config-subif)#description donskaya
sch-sochi-agko-gw-1(config-subif)#exit
sch-sochi-agko-gw-1(config)#exit
sch-sochi-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
```

Рисунок 2.6: Настройка интерфейсов маршрутизатора sch-sochi-agko-gw-1

```
sch-sochi-agko-gw-1>enable
Password:
sch-sochi-agko-gw-1#ping 10.128.255.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.128.255.5, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

sch-sochi-agko-gw-1#ping 10.128.255.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.128.255.5, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/5/13 ms

sch-sochi-agko-gw-1#
```

Рисунок 2.7: Выполнение проверки

Следующим шагом настроим площадку 42-го квартала. Для этого настроим интерфейсы у маршрутизатора `msk-q42-agko-gw-1`, коммутатора `msk-q42-agko-sw-1`, маршрутизирующего коммутатора `msk-hostel-agko-gw-1` и коммутатора `msk-hostel-sw-1` (рис. #fig:009 – #fig:017):

```
msk-q42-agko-gw-1#enable
msk-q42-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-agko-gw-1(config)#interface f0/0
msk-q42-agko-gw-1(config-if)#no shutdown

msk-q42-agko-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
msk-q42-agko-gw-1(config)#interface f0/0.201
msk-q42-agko-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.201, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed state to up
encapsulation dot1Q 201
msk-q42-agko-gw-1(config-subif)#ip address 10.129.0.1 255.255.255.0
msk-q42-agko-gw-1(config-subif)#description q42-main
msk-q42-agko-gw-1(config-subif)#exit
msk-q42-agko-gw-1(config)#interface f1/0
msk-q42-agko-gw-1(config-if)#no shutdonw
                                      ^
% Invalid input detected at '^' marker.

msk-q42-agko-gw-1(config-if)#no shutdown

msk-q42-agko-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
exit
msk-q42-agko-gw-1(config)#interface f1/0.202
msk-q42-agko-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet1/0.202, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed state to up
encapsulation dot1Q 202
msk-q42-agko-gw-1(config-subif)#ip address 10.129.1.1 255.255.255.0
msk-q42-agko-gw-1(config-subif)#description q42-management
msk-q42-agko-gw-1(config-subif)#exit
msk-q42-agko-gw-1(config)#^Z
msk-q42-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-q42-agko-gw-1#
```

Рисунок 2.8: Настройка интерфейсов маршрутизатора msk-q42-agko-gw-1

```
Password:

msk-q42-agko-sw-1>enable
Password:
msk-q42-agko-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-agko-sw-1(config)#interface f0/24
msk-q42-agko-sw-1(config-if)#switchport mode trunk

msk-q42-agko-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
exit
msk-q42-agko-sw-1(config)#interface f0/1
msk-q42-agko-sw-1(config-if)#switchport mode access
msk-q42-agko-sw-1(config-if)#switchport access vlan 201
% Access VLAN does not exist. Creating vlan 201
msk-q42-agko-sw-1(config-if)#exit
msk-q42-agko-sw-1(config)#interface f0/1
msk-q42-agko-sw-1(config-if)#switchport access vlan 201
msk-q42-agko-sw-1(config-if)#exit
msk-q42-agko-sw-1(config)#vlan 201
msk-q42-agko-sw-1(config-vlan)#name q42-main
msk-q42-agko-sw-1(config-vlan)#exit
msk-q42-agko-sw-1(config)#interface vlan201
msk-q42-agko-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan201, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan201, changed state to up
no shutdown
msk-q42-agko-sw-1(config-if)#exit
msk-q42-agko-sw-1(config)#^Z
msk-q42-agko-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-q42-agko-sw-1#
```

Рисунок 2.9: Настройка интерфейсов коммутатора msk-q42-agko-sw-1

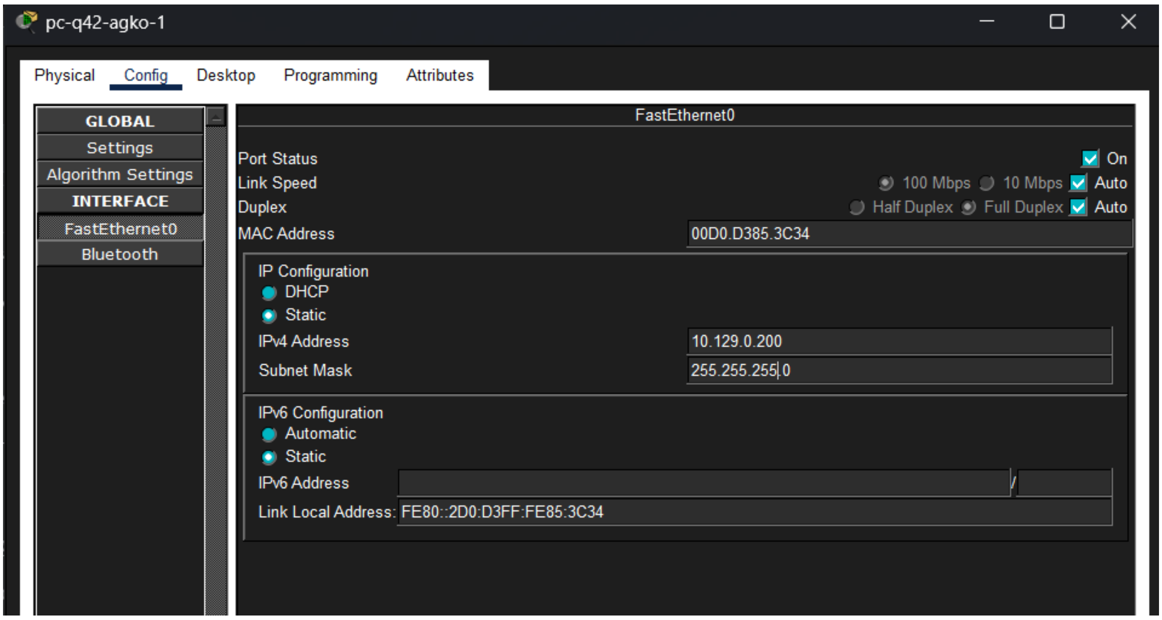


Рисунок 2.10: Присвоение адресов оконечному устройству pc-q42-1

```
C:\>ping 10.129.0.1

Pinging 10.129.0.1 with 32 bytes of data:

Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.129.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.0.1

Pinging 10.128.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.128.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Рисунок 2.11: Выполнение проверки

```
User Access Verification

Password:

msk-hostel-agko-gw-1>enable
Password:
msk-hostel-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-agko-gw-1(config)#interface g0/1
msk-hostel-agko-gw-1(config-if)#switchport trunk encapsulation dot1q
msk-hostel-agko-gw-1(config-if)#switchport mode trunk
msk-hostel-agko-gw-1(config-if)#exit
msk-hostel-agko-gw-1(config)#interface f0/1
msk-hostel-agko-gw-1(config-if)#switchport trunk encapsulation dot1q
msk-hostel-agko-gw-1(config-if)#switchport mode trunk
msk-hostel-agko-gw-1(config-if)#exit
msk-hostel-agko-gw-1(config)#vlan 202
msk-hostel-agko-gw-1(config-vlan)#name q42-management
msk-hostel-agko-gw-1(config-vlan)#exit
msk-hostel-agko-gw-1(config)#vlan 301
msk-hostel-agko-gw-1(config-vlan)#name hostel-main
msk-hostel-agko-gw-1(config-vlan)#exit
msk-hostel-agko-gw-1(config)#interface vlan202
msk-hostel-agko-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan202, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up
no shutdown
msk-hostel-agko-gw-1(config-if)#ip address 10.129.1.2 255.255.255.0
msk-hostel-agko-gw-1(config-if)#exit
msk-hostel-agko-gw-1(config)#interface vlan301
msk-hostel-agko-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up
no shutdown
msk-hostel-agko-gw-1(config-if)#ip address 10.129.128.1 255.255.255.0
msk-hostel-agko-gw-1(config-if)#exit
msk-hostel-agko-gw-1(config)#^Z
msk-hostel-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-hostel-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-agko-gw-1(config)#ip routing
msk-hostel-agko-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.129.1.1
msk-hostel-agko-gw-1(config)#^Z
msk-hostel-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-hostel-agko-gw-1#
```

Рисунок 2.12: Настройка интерфейсов маршрутизирующего коммутатора msk-hostel-agko-gw-1

```
msk-hostel-agko-gw-1#ping 10.129.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.129.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/4 ms

msk-hostel-agko-gw-1#
```

Рисунок 2.13: Выполнение проверки

```
msk-hostel-agko-sw-1>enable
Password:
msk-hostel-agko-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-agko-sw-1(config)#interface g0/1
msk-hostel-agko-sw-1(config-if)#switchport mode trunk
msk-hostel-agko-sw-1(config-if)#exit
msk-hostel-agko-sw-1(config)#interface f0/1
msk-hostel-agko-sw-1(config-if)#switchport mode access
msk-hostel-agko-sw-1(config-if)#switchport access vlan 301
% Access VLAN does not exist. Creating vlan 301
msk-hostel-agko-sw-1(config-if)#switchport access vlan 301
msk-hostel-agko-sw-1(config-if)#exit
msk-hostel-agko-sw-1(config)#vlan 301
msk-hostel-agko-sw-1(config-vlan)#name hostel-main
msk-hostel-agko-sw-1(config-vlan)#exit
msk-hostel-agko-sw-1(config)#interface vlan301
msk-hostel-agko-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up
no shutdown
msk-hostel-agko-sw-1(config-if)#exit
msk-hostel-agko-sw-1(config)#^Z
msk-hostel-agko-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-hostel-agko-sw-1#
```

Рисунок 2.14: Настройка интерфейсов коммутатора msk-hostel-sw-1

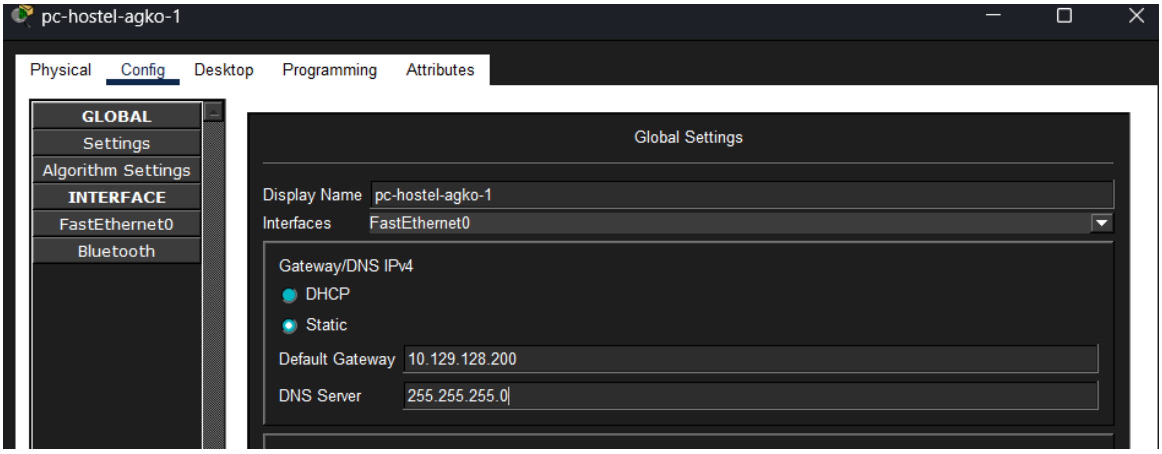
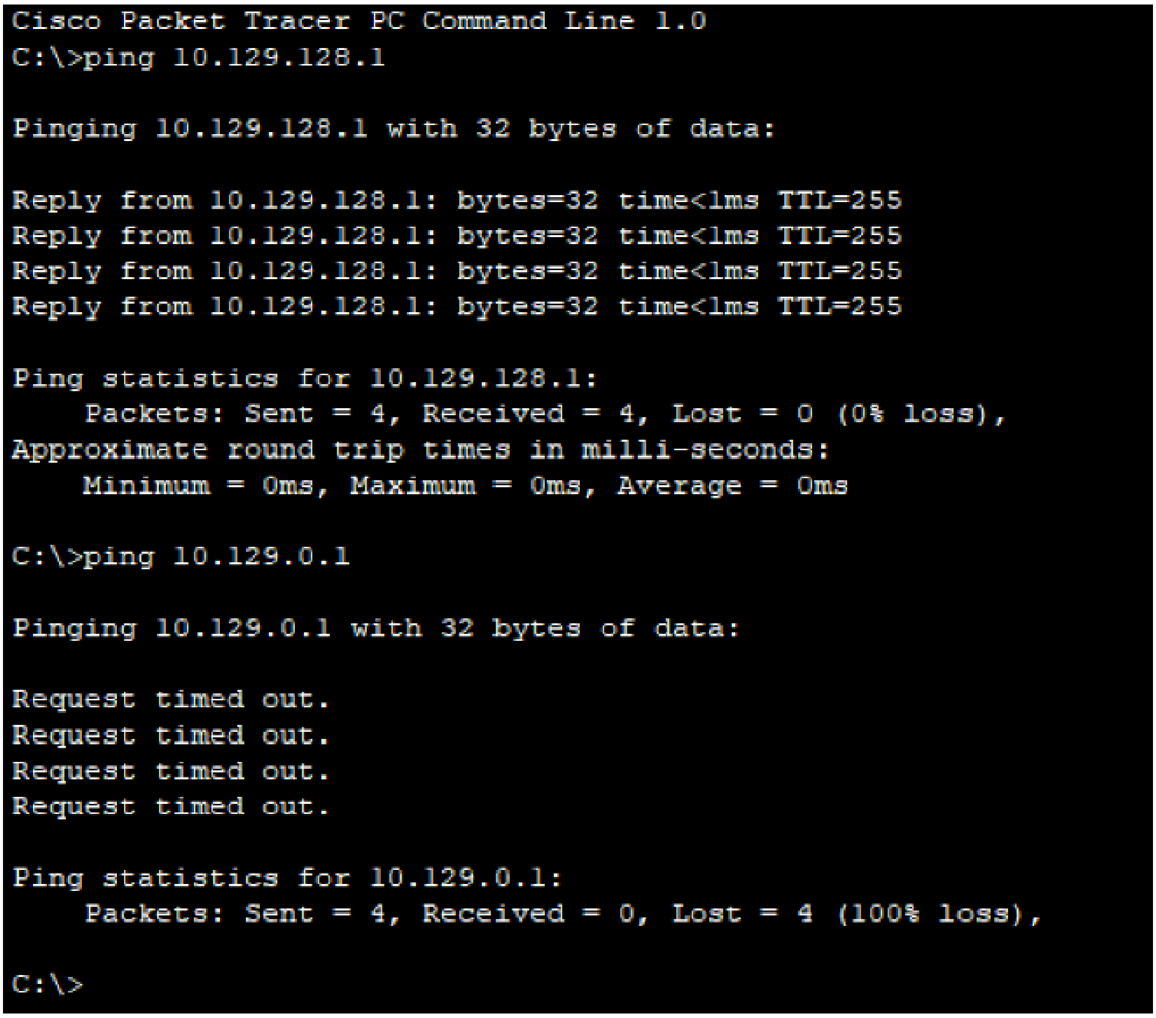


Рисунок 2.15: Присвоение адресов оконечному устройству pc-hostel-1



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.129.128.1

Pinging 10.129.128.1 with 32 bytes of data:

Reply from 10.129.128.1: bytes=32 time<1ms TTL=255
Reply from 10.129.128.1: bytes=32 time<1ms TTL=255
Reply from 10.129.128.1: bytes=32 time<1ms TTL=255
Reply from 10.129.128.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.129.128.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.129.0.1

Pinging 10.129.0.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.129.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Рисунок 2.16: Выполнение проверки

Далее настроим площадку в Сочи. Настроим интерфейсы у маршрутизатора `sch-sochi-agko-gw-1` и у коммутатора `sch-sochi-agko-sw-1` (рис. #fig:018 – #fig:020):

```
sch-sochi-agko-gw-1#
sch-sochi-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-agko-gw-1(config)#interface f0/0.401
sch-sochi-agko-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up
encapsulation dot1Q 401
sch-sochi-agko-gw-1(config-subif)#ip address 10.130.0.1 255.255.255.0
sch-sochi-agko-gw-1(config-subif)#description sochi-main
sch-sochi-agko-gw-1(config-subif)#exit
sch-sochi-agko-gw-1(config)#interface f0/0.402
sch-sochi-agko-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.402, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402, changed state to up
encapsulation dot1Q 402
sch-sochi-agko-gw-1(config-subif)#ip address 10.130.1.1 255.255.255.0
sch-sochi-agko-gw-1(config-subif)#description sochi-management
sch-sochi-agko-gw-1(config-subif)#exit
sch-sochi-agko-gw-1(config)#wr m
                            ^
% Invalid input detected at '^' marker.

sch-sochi-agko-gw-1(config)#^Z
sch-sochi-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
sch-sochi-agko-gw-1#
sch-sochi-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-agko-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.5
sch-sochi-agko-gw-1(config)#^Z
sch-sochi-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
sch-sochi-agko-gw-1#
```

Рисунок 2.17: Первоначальная настройка маршрутизатора sch-sochi-agko-gw-1

```
User Access Verification

Password:

sch-sochi-agko-sw-1>enable
Password:
sch-sochi-agko-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-agko-sw-1(config)#interface f0/1
sch-sochi-agko-sw-1(config-if)#switchport mode access
sch-sochi-agko-sw-1(config-if)#switchport access vlan 401
% Access VLAN does not exist. Creating vlan 401
sch-sochi-agko-sw-1(config-if)#switchport access vlan 401
sch-sochi-agko-sw-1(config-if)#exit
sch-sochi-agko-sw-1(config)#interface vlan 401
sch-sochi-agko-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up
exit
sch-sochi-agko-sw-1(config)#vlan 401
sch-sochi-agko-sw-1(config-vlan)#name sochi-main
sch-sochi-agko-sw-1(config-vlan)#exit
sch-sochi-agko-sw-1(config)#interface vlan401
sch-sochi-agko-sw-1(config-if)#no shutdown
sch-sochi-agko-sw-1(config-if)#exit
sch-sochi-agko-sw-1(config)#^Z
sch-sochi-agko-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
sch-sochi-agko-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-agko-sw-1(config)#interface f0/1
sch-sochi-agko-sw-1(config-if)#switchport access vlan 401
sch-sochi-agko-sw-1(config-if)#^Z
sch-sochi-agko-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
sch-sochi-agko-sw-1#
```

Рисунок 2.18: Первоначальная настройка коммутатора sch-sochi-agko-sw-1

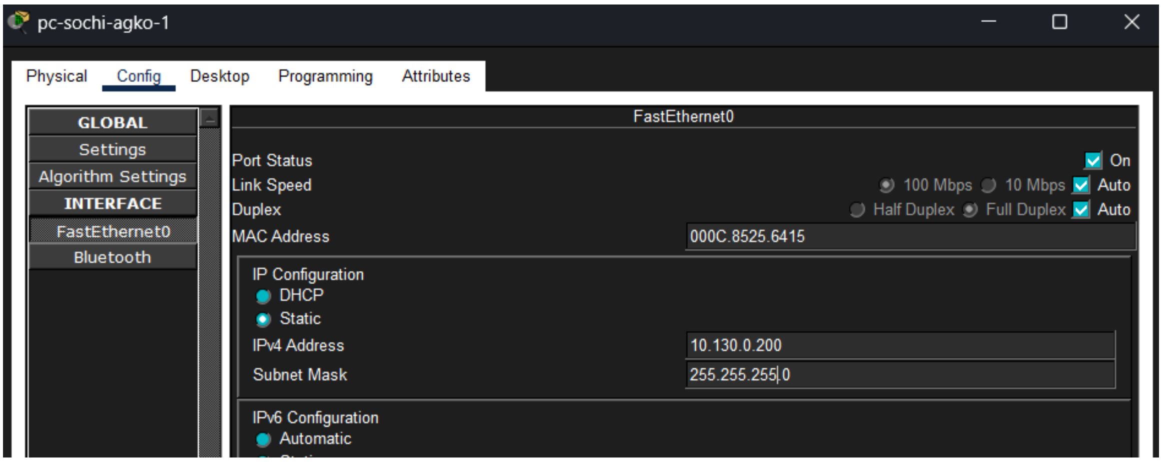


Рисунок 2.19: Присвоение адресов оконечному устройству pc-sochi-1

Затем настроим маршрутизацию между площадками. Настроим маршрутизатор `msk-donskaya-agko-gw-1`, маршрутизатор `msk-q42-agko-gw-1` и маршрутизатор `sch-sochi-agko-gw-1` (рис. #fig:021 – #fig:025):

```
msk-donskaya-agko-gw-1>enable
Password:
msk-donskaya-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-agko-gw-1(config)#ip route 10.129.0.0
% Incomplete command.
msk-donskaya-agko-gw-1(config)#ip route 10.129.0.0 255.255.0.0 10.128.255.2
msk-donskaya-agko-gw-1(config)#ip route 10.130.0.0 255.255.0.0 10.128.255.6
msk-donskaya-agko-gw-1(config)#^Z
msk-donskaya-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
```

Рисунок 2.20: Настройка маршрутизатора msk-donskaya-agko-gw-1

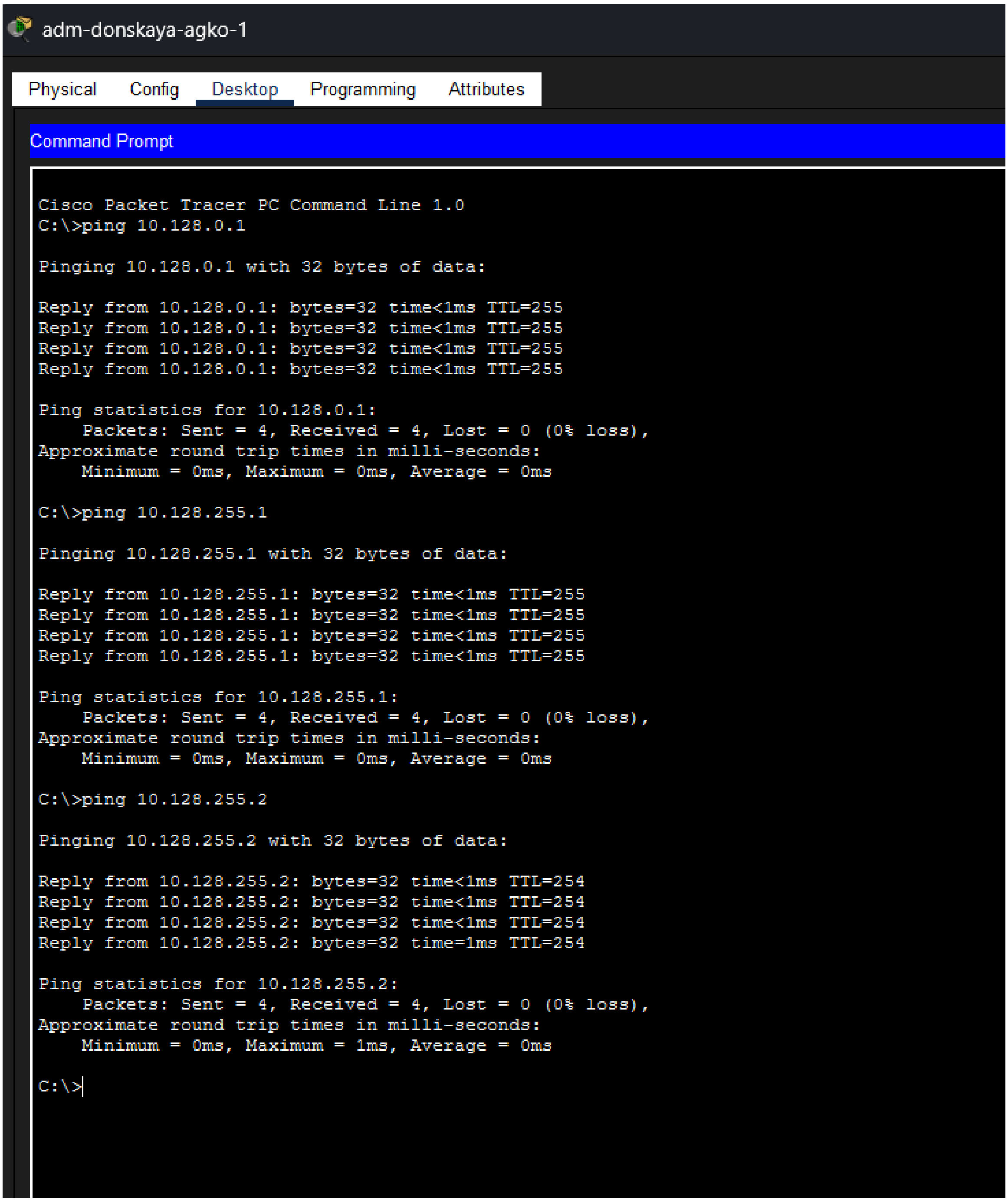


Рисунок 2.21: Выполнение проверки

```
msk-q42-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-agko-gw-1(config)#ip route 10.129.128.0 255.255.128.0 10.129.1.2
msk-q42-agko-gw-1(config)#^Z
msk-q42-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
```

Рисунок 2.22: Настройка маршрутизатора msk-q42-agko-gw-1

```
C:\>ping 10.128.255.2

Pinging 10.128.255.2 with 32 bytes of data:

Reply from 10.128.255.2: bytes=32 time<1ms TTL=254
Reply from 10.128.255.2: bytes=32 time<1ms TTL=254
Reply from 10.128.255.2: bytes=32 time<1ms TTL=254
Reply from 10.128.255.2: bytes=32 time=2ms TTL=254

Ping statistics for 10.128.255.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 10.129.0.1

Pinging 10.129.0.1 with 32 bytes of data:

Reply from 10.129.0.1: bytes=32 time=4ms TTL=254
Reply from 10.129.0.1: bytes=32 time=9ms TTL=254
Reply from 10.129.0.1: bytes=32 time<1ms TTL=254
Reply from 10.129.0.1: bytes=32 time<1ms TTL=254

Ping statistics for 10.129.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 3ms
```

Рисунок 2.23: Выполнение проверки

```
sch-sochi-agko-gw-1#
sch-sochi-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-agko-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.5
sch-sochi-agko-gw-1(config)#^Z
sch-sochi-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
sch-sochi-agko-gw-1#
```

Рисунок 2.24: Настройка маршрутизатора sch-sochi-agko-gw-1

Предпоследним шагом настроим маршрутизацию на 42-м квартале. Для этого настроим маршрутизатор `msk-q42-agko-gw-1` (рис. #fig:026) и маршрутизирующий коммутатор `msk-hostel-agko-gw-1` (рис. #fig:027):

```
msk-q42-agko-gw-1>enable
Password:
msk-q42-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-agko-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.1
msk-q42-agko-gw-1(config)#^Z
msk-q42-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-q42-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-agko-gw-1(config)#ip route 10.129.128.0 255.255.128.0 10.129.1.2
msk-q42-agko-gw-1(config)#^Z
msk-q42-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
```

Рисунок 2.25: Настройка маршрутизатора msk-q42-agko-gw-1

```
msk-hostel-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-agko-gw-1(config)#ip routing
msk-hostel-agko-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.129.1.1
msk-hostel-agko-gw-1(config)#^Z
msk-hostel-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-hostel-agko-gw-1#
```

Рисунок 2.26: Настройка интерфейсов маршрутизирующего коммутатора msk-hostel-agko-gw-1

И наконец последним шагом настроим NAT на маршрутизаторе `msk-donskaya-agko-gw-1` (рис. #fig:028) и выполним контрольную проверку (рис. #fig:029):

```
msk-donskaya-agko-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-agko-gw-1(config)#interface f0/1.5
msk-donskaya-agko-gw-1(config-subif)#ip nat inside
msk-donskaya-agko-gw-1(config-subif)#exit
msk-donskaya-agko-gw-1(config)#interface f0/1.6
msk-donskaya-agko-gw-1(config-subif)#ip nat inside
msk-donskaya-agko-gw-1(config-subif)#exit
msk-donskaya-agko-gw-1(config)#ip access-list extended nat-inet
msk-donskaya-agko-gw-1(config-ext-nacl)#remark q42
msk-donskaya-agko-gw-1(config-ext-nacl)#permit ip host 10.129.0.200 any
msk-donskaya-agko-gw-1(config-ext-nacl)#permit ip host 10.129.128.200 any
msk-donskaya-agko-gw-1(config-ext-nacl)#remark sochi
msk-donskaya-agko-gw-1(config-ext-nacl)#permit ip host 10.130.0.200 any
msk-donskaya-agko-gw-1(config-ext-nacl)#exit
msk-donskaya-agko-gw-1(config)#^Z
msk-donskaya-agko-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-donskaya-agko-gw-1#
```

Рисунок 2.27: Настройка NAT на маршрутизаторе msk-donskaya-agko-gw-1

```
C:\>ping 192.0.2.1

Pinging 192.0.2.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.0.2.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.129.0.1

Pinging 10.129.0.1 with 32 bytes of data:

Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time=9ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255
Reply from 10.129.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.129.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>
```

Рисунок 2.28: Контрольная проверка

3 Вывод

В ходе выполнения лабораторной работы мы настроили взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи.

4 Ответы на контрольные вопросы

1. **Приведите пример настройки статической маршрутизации между двумя подсетями организации.**

   Необходимо задать IP шлюзов на интерфейсах, настроить sub-интерфейсы с тегированием кадров VLAN и своими IP, затем настроить статические маршруты между сетями.

2. **Опишите процесс обращения устройства из одного VLAN к устройству из другого VLAN.**

   1-е устройство посылает фрейм на маршрутизатор, тот меняет MAC источника на свой и перенаправляет фрейм 2-му устройству.

3. **Как проверить работоспособность маршрута?**

   `ping` на диаметрально противоположных устройствах друг к другу.

4. **Как посмотреть таблицу маршрутизации?**

   `show ip route`